

Christopher Musco

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Academic Positions

New York University <i>Assistant Professor, Tandon School of Engineering</i> <i>Department of Computer Science and Engineering</i>	New York, NY <i>2019 - present</i>
Microsoft Research <i>Visiting Researcher, Machine Learning and Optimization Group</i>	Redmond, WA <i>Summer 2020</i>
Princeton University <i>Research Instructor, Department of Computer Science</i>	Princeton, NJ <i>2018 – 2019</i>
Massachusetts Institute of Technology <i>Research and Teaching Assistant, Theory of Computation Group</i>	Cambridge, MA <i>2013 – 2018</i>
Yahoo Labs <i>Research Intern, Scalable Machine Learning Group</i>	New York, NY <i>Summer 2015</i>

Research Areas

foundations of data science • scalable machine learning • numerical linear algebra • theory of algorithms
randomization • dimensionality reduction • sketching & streaming • spectral graph algorithms • optimization

Education

Massachusetts Institute of Technology <i>Ph.D. in Computer Science</i> Advisor: Jonathan Kelner Thesis: Faster Linear Algebra for Data Analysis and Machine Learning M.S. Thesis: Dimensionality Reduction for Sparse and Structured Matrices	Cambridge, MA <i>2013 – 2018</i>
Yale University <i>B.S. in Applied Mathematics, B.S. in Computer Science</i>	New Haven, CT <i>2008 – 2012</i>

Publications

*Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science. Any exceptions are marked with a *, which indicates that authors are ordered by contribution.*

Linear and Sublinear Time Spectral Density Estimation. Vladimir Braverman, Aditya Krishnan, Christopher Musco. *Preprint* 2021.

Dynamic Trace Estimation. Prathamesh Dharangutte, Christopher Musco. *Preprint* 2021.

Correlation Sketches for Approximate Join-Correlation Queries*. Aécio Santos, Aline Bessa, Fernando Chirigati, Christopher Musco, Juliana Freire. *ACM Special Interest Group on Management of Data Conference (SIGMOD)* 2021.

Public Transport Planning: When Transit Network Connectivity Meets Commuting Demand*. Sheng Wang, Yuan Sun, Christopher Musco, Zhifeng Bao. *ACM Special Interest Group on Management of Data Conference (SIGMOD)* 2021.

Simple Heuristics Yield Provable Algorithms for Masked Low Rank Approximation. Cameron Musco, Christopher Musco, David Woodruff. *Innovations in Theoretical Computer Science (ITCS)* 2021.

Hutch++: Optimal Stochastic Trace Estimation. Raphael Meyer, Cameron Musco, Christopher Musco, David P. Woodruff. *SIAM Symposium on Simplicity in Algorithms (SOSA)* 2021.

Graph Learning for Inverse Landscape Genetics. Prathamesh Dharangutte, Christopher Musco. *AAAI Conference on Artificial Intelligence (AAAI)* 2021. Short version in NeurIPS 2020 AI for Earth Sciences workshop.

Fourier Sparse Leverage Scores and Approximate Kernel Learning. Tamás Erdélyi, Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2020. Invited for spotlight presentation (4% of submissions).

Near Optimal Linear Algebra in the Online and Sliding Window Models. Vladimir Braverman, Petros Drineas, Cameron Musco, Christopher Musco, Jalaj Upadhyay, David P. Woodruff, Samson Zhou. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2020.

The Statistical Cost of Robust Kernel Hyperparameter Turning. Raphael A. Meyer, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2020.

Finding the Mode of a Kernel Density Estimate. Jasper C.H. Lee, Jerry Li, Christopher Musco, Jeff M. Phillips, Wai Ming Tai. *In submission*. 2020.

Low-Rank Toeplitz Matrix Estimation via Random Ultra-Sparse Rulers. Hannah Lawrence, Jerry Li, Cameron Musco, Christopher Musco. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. 2020.

Sample Efficient Toeplitz Covariance Estimation. Yonina Eldar, Jerry Li, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

Fast and Space Efficient Spectral Sparsification in Dynamic Streams. Michael Kapralov, Aida Mousavifar, Cameron Musco, Christopher Musco, Navid Nouri, Aaron Sidford, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

Understanding Filter Bubbles and Polarization in Social Networks. Uthsav Chitra, Christopher Musco. *ACM International Conference on Web Search and Data Mining (WSDM)* 2020. Preliminary version in KDD Workshop on Issues of Sentiment Discovery and Opinion Mining.

A Universal Sampling Method for Reconstructing Signals with Simple Fourier Transforms. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *ACM Symposium on Theory of Computing (STOC)* 2019.

Learning Networks from Random Walk-Based Node Similarities. Jeremy Hoskins, Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *Conference on Neural Information Processing Systems (NeurIPS)* 2018.

Eigenvector Computation and Community Detection in Asynchronous Gossip Models. Frederik Mallmann-Trenn, Cameron Musco, Christopher Musco. *International Colloquium on Automata, Languages, and Programming (ICALP)* 2018.

Minimizing Polarization and Disagreement in Social Networks. Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *The Web Conference (WWW)* 2018.

Stability of the Lanczos Method for Matrix Function Approximation. Cameron Musco, Christopher Musco, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2018.

Recursive Sampling for the Nyström Method. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2017.

Random Fourier Features for Kernel Ridge Regression: Approximation Bounds and Statistical Guarantees. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *International Conference on Machine Learning (ICML)* 2017.

Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling. Michael B. Cohen, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2017.

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *SIAM Algorithm Engineering and Experiments (ALENEX)* 2017.

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *International Conference on Machine Learning (ICML)* 2016.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2015. Invited for full oral presentation (1 of 15, < 1% of submissions). Also presented at the 2016 Copper Mountain Conference on Iterative Methods

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *ACM Symposium on Theory of Computing (STOC)* 2015.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiroglou-Douskos, Martin Rinard. *Network and Distributed System Security Symposium (NDSS)* 2015.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *Innovations in Theoretical Computer Science (ITCS)* 2015.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2014, Special Issue SIAM Journal on Computing 2017.

Student and Postdoc Advising

Aline Bessa. Postdoc, New York University.	2021 - present
Apoorv Singh. Ph.D., New York University.	2020 - present
Raphael Meyer. Ph.D., New York University.	2019 - present
Indu Ramesh. M.S., New York University.	2020 - present
Xinyu Luo. M.S., New York University.	2020 - present
Prathamesh Dharangutte. M.S., New York University.	2019 - present
John Shin. M.S., New York University.	2019 - 2020
Tristan Pollner. Undergraduate, Princeton.	2018 - 2019

Funding

NSF CAREER Award

Frontiers in Matrix Sketching, Sole PI, \$562,224. 2021-2026

Teaching

Algorithmic Machine Learning and Data Science (NYU CS-GY 9223I)

Course Instructor

Fall 2020, Fall 2019

Introduction to Machine Learning (NYU CS-UY 4563)

Course Instructor

Spring 2020

Advanced Algorithm Design (Princeton COS 521)*Course Instructor**Fall 2018***Advanced Algorithms (MIT 6.854/18.415)***Teaching Assistant**Spring 2016***Technical Communication Skills for Graduate Students (MIT 6.S977)***Workshop Leader**Spring 2016***Mentorship and Outreach**

SIAM/MathWorks Math Modeling Challenge*Problem Development Committee, Judge**2018 – present*

Support SIAM's annual math modeling competition, which gives high school students an opportunity to apply mathematics, statistics, and data science to a real-world problem. Serve on committee in charge of writing practice and contest problems. Score and provide feedback on student submissions to the contest.

MIT Graduate Communication Lab*Communication Adviser**2015 – 2018*

Helped to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offered feedback, guidance, and coaching to individual students during weekly office hours.

CSAIL Algorithms Office Hours*Member**2016 – 2018*

Advised researchers, generally in applied fields, on framing and solving algorithmic problems arising in their work.

Research Science Institute at MIT*Summer Research Adviser**2014*

Primary supervisor for a high school student participating in the prestigious RSI summer research program. Advisee's project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Talks and Presentations

Sublinear Time Spectral Density Estimation

Cornell University, Scientific Computing and Numerics Seminar

*Mar. 2021***Importance Sampling for Approximating High-Dimensional Kernel Matrices**

Emory University, Numerical Analysis and Scientific Computing Seminar

*Feb. 2021***Optimal Stochastic Trace Estimation**

Stanford University, Operations Research Seminar

*Oct. 2020***Introduction to Machine Learning**

New York University, guest lecture in "Introduction to Computer Science"

*Mar. 2020***Structured Covariance Estimation**

Yahoo! Research, Seminar

April 2020

Johns Hopkins University, Theory Seminar

Oct. 2019

Symposium on Discrete Algorithms (SODA)

*Jan. 2020***From Randomized Linear Algebra to Randomized Functional Analysis**

SIAM Mathematics of Data Science Conference

May 2020

Rutgers University, DIMACS RandNLA Workshop

Sept. 2019

University of Utah, Data Science Seminar

*Jan. 2020***Learning Signals with Simple Fourier transforms**

Symposium on Theory of Computing (STOC)

June 2019

Columbia University, Theory Seminar	<i>Oct. 2018</i>
UC Berkeley, Simons Institute Workshop	<i>Sept. 2018</i>
Princeton University, Theory Seminar	<i>Sept. 2018</i>
The Lanczos Method in Data Science: Challenges and Opportunities	
SIAM Conference on Parallel Processing for Scientific Computing, Minisymposium	<i>Feb. 2020</i>
Georgetown University, CS Seminar	<i>Feb. 2018</i>
Symposium on Discrete Algorithms (SODA)	<i>Jan. 2018</i>
Unifying Random Features and Leverage Scores for Kernel Matrix Approximation	
Institute for Advanced Study, Computer Science and Discrete Mathematics Seminar	<i>Feb. 2018</i>
Harvard University, guest lecture in "Sketching Algorithms for Big Data"	<i>Nov. 2017</i>
Microsoft Research, Redmond	<i>Sept. 2017</i>
Sketching and Sampling Methods for Large Scale Linear Algebra	
Twitter, Cambridge	<i>July 2017</i>
Recursive Sampling for the Nyström Method	
Neural Information Processing Systems (NeurIPS) poster session	<i>Dec. 2017</i>
New England Machine Learning Day poster session	<i>May 2017</i>
Algorithms for Determining Tournament Payout Structures	
Meeting on Algorithm Engineering and Experiments (ALENEX)	<i>Jan. 2017</i>
Introduction to Linear Sketching	
MIT, Theory Retreat	<i>Sept. 2016</i>
Principal Component Projection without Principal Component Analysis	
International Conference on Machine Learning (ICML)	<i>June 2016</i>
Ridge Leverage Score Sampling	
National Institute of Informatics, Shonan Meeting	<i>July 2016</i>
MIT, Algorithms & Complexity Seminar	<i>July 2016</i>
University of Utah, Data Group Meeting	<i>Jan. 2016</i>
Randomized Block Krylov Methods	
Neural Information Processing Systems (NeurIPS) poster session	<i>Dec. 2015</i>
Large Scale Column Subset Selection	
Yahoo Labs, Science Week poster session	<i>July 2015</i>
Dimensionality Reduction for k-Means Clustering	
IBM T.J. Watson, Mathematical Sciences research seminar	<i>Aug. 2015</i>
Symposium on Theory of Computing (STOC)	<i>June 2015</i>
MIT, Theory Lunch	<i>June 2014</i>
Principled Sampling for Anomaly Detection	
MIT, Defense Advanced Research Projects Agency site visit	<i>Mar. 2015</i>
Network and Distributed System Security Symposium (NDSS)	<i>Feb. 2015</i>
Uniform Sampling for Matrix Approximation	
Yahoo Labs, Scalable Machine Learning research seminar	<i>June 2015</i>
Single Pass Spectral Sparsification in Dynamic Streams	
MIT, Annual Sublinear Algorithms Day poster session	<i>Apr. 2015</i>

Harvard University, Theory Seminar	Nov. 2014
Foundations of Computer Science (FOCS)	Oct. 2014

Service

Organization.....	
SIAM Mathematics of Data Science Minisymposium Organizer	2020
NYU Tandon Algorithmic Data Science Reading Group	2019 - present
Department Service.....	
Faculty Search Committee	2019,2021
Online Faculty Oversight Committee	2021 - present
Ph.D. Admissions Committee	2019 - present
Conference Programming.....	
NeurIPS Area Chair	2019
Conference Reviewing.....	
STOC, FOCS, SODA, ICALP, ITCS, ICML, NeurIPS, AAAI, ESA, APPROX, RANDOM, STACS, IWOCA, WALCOM, NDSS	
Journal Reviewing.....	
ACM Transactions on Algorithms, Journal of Machine Learning Research, SIAM Journal on Matrix Analysis and Applications, SIAM Journal on Scientific Computing, SIAM Journal on Mathematics of Data Science, Science Advances, Mathematical Programming, IEEE Transactions on Signal Processing, IEEE Transactions on Knowledge and Data Engineering, International Journal of Foundations of Computer Science, Journal on Selected Areas in Information Theory	
Grants Panels.....	
NSF Division of Information and Intelligent Systems (2019)	
Notables.....	
ICML 2018 Outstanding Reviewer, ICML Expert Review 2021	

Invited Workshops and Events

Rutgers, DIMACS	
<i>Invited participant</i>	2019
Workshop on Randomized Numerical Linear Algebra, Statistics, and Optimization.	
UC Berkeley, Simons Institute for the Theory of Computing	
<i>Invited participant</i>	2018
Workshop on Randomized Numerical Linear Algebra and Applications.	
Japan National Institute of Informatics	
<i>Invited participant</i>	2016
Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra.	

Honors and Awards

National Science Foundation	2014-2017
<i>Graduate Research Fellowship, recipient.</i>	
Yale University	2012
<i>Cum Laude, Distinction in Both Majors.</i>	

Other Professional Experience

Redfin

Software Engineer, Data Team

Seattle, WA

2012 – 2014

Elysium Digital

Summer Consultant

Cambridge, MA

2011

Amicus

Software Developer

New Haven, CT

2010 – 2011